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H8SCOOP

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BITS & PIECES

This December issue is being mailed early as mentioned last month to avoid the Christmas mail hassels! Your January 1982 issue will be mailed early January, and the February 1982 issue will be mailed late January or early February, and after that, all should be on schedule. Since I will have no December rush, I plan to do some vacationing, so if you want to call and can't reach me, it's because I'm on vacation. I'll try to have my phone calls forwarded so you can get some information about my return time.

This will be a larger than normal issue, since I have a backlog of information to publish, I felt I should take this time to give you a bigger issue--that's right, at no extra cost to you. I will be sending the yearly index out in this issue or next, depending on space. To use this, items are classified according to listing classification in H8SCOOP or GENERAL ITEM. Each has a number such as 12-4 which means issue 12 page 4. It's as simple as that. Keep in mind, that this is a general index. Obviously I have not included all items, just items I felt would be important to want to look up in the future. Although it's not the greatest index, it's much better than nothing.

IS CASSETTE DEAD?

I receive requests from time to time on software and tips for cassette tape users. I well realize most of us have gone to the disk system, but not all. I also realize that this newsletter has published very little information about the cassette system in the past year. The reason for this is nothing is being submitted. I am at a loss to "fill in" material on cassette usage since I no longer run the system, and depend on input from subscribers for this. If anyone is running the cassette system and has tips, information, or software, please send me information so we can do something for those less fortunate out there who cannot spend the big bucks on disk systems. That's the only way I will have anything to print--no input, no output!

H89 CHIP SET ROM

Grant Gustafson writes in "Many people beleive that the \$55 chip set, is needed to expand memory or run CP/M on the H89. Those who read documentation will see the need for P/N 444-66, which is the \$11.95 chip and the \$1.00 brown cable 134-1120-00. This leaves the tape board intact, and load/dump operates. CP/M 2.2, 1.43, and 1.42 run on this, from 48K to 64K and HDOS runs 48-56K. The two extra chips? Yes, you need them for the H47 drive, but who's got \$3200?

SPELLBINDER KEYTOPS

I now have a set of custom engraved SPELLBINDER keytops for my H89 to help me better when I use Spellbinder. These keys are red and blue and can be installed on the H19 or H89 in about half an hour. They give the terminal a very professional look and help make more sense out of the functions of Spellbinder. I feel these keys used with Spellbinder make this the top dog in word processors. If you are interested, they cost \$25. plus shipping from ARKAY ENGRAVERS INC, POB 916, Bellmore, NY 11710 (516) 781-9859. In addition to these, they will engrave anything you want on your keytops for \$1.02 each key.

If you get the Spellbinder set, here's where they go. INDENT is f1, SOFT HYPHEN is f2, MODE ENHANCE is f3, ENTER ENHANCE is f4, PRINT CHAR is blue square, MARK is red square, REWRITE is white square, CLEAR is line feed, DELETE TEXT is delete. These are all red. On

the keypad, blue CURSOR MODE is 7, red UP ARROW is 8, MODE DELETE (blue) is 9, LEFT ARROW (red) is 4, CURSOR SCAN (red) is home, RIGHT ARROW (red) is 6, MODE BKWRD (blue) is 1, DOWN ARROW (red) is 2, MODE FWARD (blue) is 3, EDIT/COMM (red) is 0, and INSERT (red) is enter.

NEWSLETTER FOR FARMERS

Lloyd Dinkins/ POB 12285/ Memphis, TN 38112/ (901) 274-9030 plans to publish a newsletter starting in January 1982, specifically for farmers and their computers. He will offer programmers a good market if they can come up with some BASIC programs for farmers to use in their management. Lloyd says if some really good software writer would get interested in agriculture, learn what farmers need, and get busy supplying it, he could make a bundle. Agriculture is far and away the largest industry in the world, but just now getting into desktop computers. A recent survey indicated about 4% of American farmers are now using micros, with a big surge expected in the near future. This is the market Lloyd plans to capture with his newsletter, Agriculture only.

While on this subject, I want to add there are a lot of Apples and TRS's being sold to farmers for just that purpose, and Lloyd is right, there's a very good future here selling systems and software to farmers. Those that want to make some bucks and have this background and/or interest, this is the way to go. The way to make money in this industry is to be first with this type of thing and get in on the ground floor. It's wide open, any takers?

DEAL FROM MI-8

Ted Benglen of Micro-Interface for the H8/H89 computers writes in to tell me he has made arrangements with Don of Environmental Systems for his PC-12/3 port board. This offer will extend through December 15th. The V-8 Model II voice synthesizer and the PC-12/3 purchased through MI-8 will cost only \$474.95. This represents a \$114.95 savings over the separately purchased price. This is an H8SCOOP exclusive. You must mention H8SCOOP when ordering, names may be verified before shipment.

Both boards come assembled and tested, ready to run. The PC-12 is a must for anyone planning to add any of MI-8's or anyone else's parallel I/O peripherals, such as the V8 or BSR, and there are more coming soon, like the temperature acquisition for the PC-12 and input voice recognition in addition to the V8. This PC-12 comes with 3 ports and can easily be upgraded to a full 12 port board as needed.

Also through the month of December Ted is offering one free heatsink extension to anyone that writes and requests one. There is a limited supply, so don't wait. This is only for the H8 boards.

Soon to be available will be a parallel/clock board for the H89. It will have user options of a real time clock and 6 parallel ports, or 9 without the clock.

Ted also wishes to thank all Heath users that have supported MI-8 this first year. Next year looks even better with MI-8 moving into a larger facility and developing an even wider line of products for the Heath computer. It's been a great first year and Ted wants to thank all out there for making it possible.

REVIVAL OF H19 19200 BAUD ISSUE

I devoted quite a bit of space last year to H19's running at

19,200 baud and thought I had seen the last of it, but information is coming in again. Lloyd Richard writes in "In your June 1980 H8SCOOP you indicate some of the earlier H19's would run at 19,200 baud but according to Heath not so on some of the newer units. I have had my H19 for about three months now and it was purchased new from Heath. So now it appears the newer H19 etc. will run at the higher rate.

Lloyd also adds this tip for running the H19/H89 at 19,200. "It is not necessary to change the JP10 jumpers on the terminal logic board in the H19/H89 in order to increase the baud rate from 9600 to 19200. Merely change the setting of SW #401 on the terminal logic board. Thus the Z80 chip would not have to be replaced with a Z80-A.

The switch setting for the 19200 (left to right) would be 1011. Turn the power on and your terminal is configured for 19,200 baud. HDOS 2.0 records this rate on the disc the first time the disc is booted, but it can be changed.

Attempt to boot the system. You may get garbage or possibly nothing on the screen. VERY QUICKLY press and release the 'break' key and type spaces until the ACTION BOOT message appears (should happen after 3 to 5 spaces). If it does not, you have held the 'break key' down too long, in which case reboot and try again. The 'break' key causes HDOS to look for spaces to determine the new baud rate. Holding the 'break' key down too long will cause breaks to be examined for baud rate instead of spaces. It is important to point out that this only works to increase the baud rate on a previously booted disc. To decrease the baud rate DO NOT use the break key, but attempt to boot and then type spaces until the 'ACTION BOOT' message appears.

Some comments on this from Bill Hall follow. "As you pointed out in your newsletter of some months ago, older terminals will run at this speed (19200), and you can access it by typing ESC r M or by setting \$401 at 1011." Bill says the same thing about the Break key, but adds "if you have modified HDOS (to time out in 1 or 2 seconds and auto boot), then use the DUMP program and patch track and sector zero, bytes 5 and 6 to the LS and MS values of the needed divisor latch values respectively. For example, if your disks are now set for 9600 baud, you will find 0C and 00 at these locations. Change them to 06 and 00 and you have 19,200 baud. For fun, try setting these to 17 and 4 respectively, type ESC r A offline on your terminal, and watch how it was done in the bad old days!

"Now let me make some remarks about using this speed. I tried it for several months and then went back to 9600 baud for a very simple reason. The terminal misses the ESC sequences used by the PIE editor from time to time. This can do damage to your files since it usually leaves the second letter of the sequence in your file. It would be interesting if other readers will report their experiences, especially if they are using PIE or other programs making extensive use of the special keys.

Comments from Lloyd Johnson. "If all of your disks have been set to 9600 Baud rate, do not change any Baud rate settings before BOOTING a disk which is not Write protected. Use DUMP to find Track 0, Sector 0 of that disk. The Baud rate code is at address 0005 of that sector. It will be 0CH for 9600 Baud. Change it to 06H for 19,200 Baud. **Do not return to Command mode.** Instead RESET and try to reBOOT. If it fails to reBOOT, the code was changed. Turn off your H/Z89 and change bit 6 of CPU switch SW501 and bit 1 of terminal switch S401 to ON, ie. from 0 to 1. The altered disk should then BOOT. Dump can be used to change the first sector of other disks by swapping them in SY0:--just be sure to RESET after each change.

Another note from Robert Quinn. "I found an easy way to change the baud rate stored on disk by doing the following:

"Boot a Ver 2.0 disk that has not been booted before, Like your original source. The first time up the disk asks for SPACE (baud rate), **don't respond.** Instead remove that disk and install the disk needing the change. Now type the SPACE BAR. Disk should come up and reflect new baud rate." Robert says he is not sure if it will work for higher baud rates.

Y CONNECTOR FOR DISK CABLE

If you are running an additional drive and don't have the proper cable arrangements, for about \$10 from JADE (or other places) you can get a 34 contact INTRA-CONNECTOR which plugs into your controller board and provides two 34 pin outlets to allow you to plug two cables for disk drives. I have installed a disk drive in my H19 to allow me to run 3 drives on my system with the 2 drive H17 and found this part to be just fine. Now I run one cable from my H8 to the H19 and one to the H17. The only negative thing about it, is with the H17 board in the H8 you need two slots worth of space to allow for the extra room this thing needs. In the H89 it works out better.

DISK CONVERSION SERVICE

I have decided to make a disk conversion service available since it is in demand and I know of no one else doing it. It is common especially among CP/M people since their software is only available on 8" disks, and this is the only time you need 8". The problem may come since you don't want to buy an 8" drive and controller for these limited applications. I will for a fee of \$10 which includes handling and postage, convert any standard CP/M disk to Heath 5.25" standard format, Magnolia DD format 5.25" or Heath 390K Tandon format, or other combinations. With this order, include the disk needing to be converted along with the necessary discs to be copied to (or allow extra for discs to be supplied). If interested write for complete price lists and conversions available.

TERMINAL TURNTABLE IDEA

Cesar Dominguez writes in an idea for a terminal turntable, often costing over \$200. He says to locate a bar or kitchen stool with a swivel seat and put the swivel between two boards, and presto, a turnable terminal. There's more to it than that, but it can be made in about two hours and under \$15. He says design refinements are easy. He recessed the swivel into the wood for less spacing. Also make note of the center of gravity for your terminal so it is centered with respect to its gravity center rather than it's physical center.

H8 COMPUTER REPAIR SERVICE

Trionyx Electronics/ POB 5131/ Santa Ana, CA 92704 is offering a repair service for H8 computers. The service will cost \$50 plus parts and return, and will be limited to CPU box (H8), which should contain all the Heath pc boards, as well as any of their boards. Their new T-H90 motherboard (which is the way to go anyway--ed) must be installed in the H8 or ordered with the repair service. This I imagine is for reliability.

The primary purpose of this service is to encourage H8 owners to use their products. The Heath company and local Heathkit centers will not service H8 computers with products obtained from outside manufacturers, so you have to have somewhere to turn. That's sad, since I don't know of hardly any H8's that are all Heath, as Heath was far to slow with goodies for the H8! Because of this, many H8ers are reluctant to purchase non-Heath products for their H8's. We also know the H8 is being 'shelved' at Heath, and it's nice to know people like Trionyx exist to continue giving us full support!

If the Trionyx mother board is not installed, it can be installed for \$250 completely assembled, or \$175 for the Heath equivalent assembly. Return shipping is an extra \$20 in USA and Canada. \$70 must be received to start any

work on the H8, and if ordered with repair, the motherboard will be assembled and installed at no additional cost beyond standard ordering cost.

H8 CPU boxes (H8's) will be returned fully functional and reliable operation guaranteed. The motherboard will be implemented to the highest level required for this purpose, on an individual basis. Parts will also be replaced on all boards, as required, to guarantee reliable operation. Figure three to four weeks to service and return an H8. All units should be sent fully insured and well packed to Trionyx Electronics Inc., 2287 S. Grand Ave., Santa Ana, CA 92705. Computer peripherals will not be serviced.

UPDATE FROM ULTIMETH CORP.

Dean Gibson of Ultimeth Corp, 24025 Fernlake Dr, Harbor City, CA 90710, wanted me to mention that an improvement has been made in the version 2 of the DKH17.DVD device driver. All new versions distributed will incorporate this improvement. Anyone who has purchased the first version (before 11/1/81) of this update can get the new version free by sending the V2 distribution disc to Dean or Myself. There have been questions as to what the V2 driver did that the V1 did not. The biggest change was the ability to recognize and automatically adjust itself for 2 or 4 Mhz operation. In addition, several bugs that were found in the Heath code were corrected or worked around. Also, the device driver slowing down with long select times has been fixed.

A note on the MTRHEX and MTROCT ROM's for the H89. Several features are available which were not mentioned in my last writeup. These are replacement monitor ROM'S for the H89 which provide increased HDOS processing speed and additional debugging capabilities. The following features have been incorporated.

The monitor clock interrupt processing routine has been rewritten, giving an effective 16% CPU speed increase to HDOS programs running on a 2MHz system, and a 7% increase at 4MHz. The clock (TICCNT) has been extended to a four byte counter. The new monitor now supports debugging of programs under HDOS. Additional monitor debugging commands have been provided to display and alter all Z-80 registers, to read and write to I/O ports, and to single step and breakpoint programs, and to restart HDOS. With an optional hardware mod, the monitor will allow the RESET key to interrupt a program.

The dynamic RAM test has been updated to provide faster and better testing, and the disk drive test now correctly measures the rotation time of any mini-floppy unit. The monitor supports CPU speeds of 2 or 4 MHz, including booting from an H17 drive, and H47 drive, and Magnolia's DD floppy controller board. (Additional software, available from Ultimeth Corp., may be required). Full compatibility with HDOS and Heath and Magnolia CP/M has been maintained. Each ROM retains all of the common Heath 89 ROM entry point locations except for cassette tape processing.

The monitor ROM is available in split octal (MTROCT) or hexadecimal (MTRHEX) for \$50, specify version. It is a direct plug replacement for the Heath ROM (TMS 2716 format) unless requested in 5V 2716 format which requires moving 3 jumpers on the CPU board. Previous purchasers of UltiMeth ROMs may exchange their old ROMs for new ones at no charge.

FEEDBACK ON 'ETHICS' EDITORIAL

I have received quite a bit of feedback on my last editorial, **ETHICS**. Two against and the rest for. One point against said "I have been stuck with some stuff which isn't worth the powder to blow it to you know where. As a result, I will not buy any software until I try it first. Now you jolly well know that no software vendors are going to send anything for review and there is no known way to get your money back for a piece of junk."

"Yes, I have some software of unknown parentage and I have traded for some with the same circumstances. Were it not for this fact I would not have gotten interested in computers and I know I would not have learned as much without some programs to play with.. This is the other side of my coin."

So we see, there're two ways to look at something. I must say, I have to agree in the sense that I have also been 'stuck' with some worthless junk. One purpose of H8SCOOP is to try to prevent that with GOOD GUYS & BAD GUYS columns, REPORT columns, and NEW STUFF. I try to bring out what I know first hand and what I have heard from others. If I feel something is good enough (like Spellbinder) I decide not only to handle it myself, but to make the price the best around. This I do, not to get rich, but to help my readers out there get the best and most for their money. I also feel some software authors are going overboard. I purchased MUNCHKINS the other day (it's fantastic!) and the info with it said something like licensed for use on a single system only. That's going too far. I have 3 systems up and running here, and that would mean I have to buy three copies if I were going to run it on each system. Come on! If one user has 20 systems, it should be OK to run it on each and any of them!

Back to the point, it is unfortunate there is junk out there, but I'm sure it's not intentional. Some of the stuff I have got is business software that evidently has worked in a specific application. The guy is so excited about it, he markets it, with little or no concept of the "real world" and "different applications" besides his exact application, and that's where the problems start. Let's face it, when you buy a business package for \$50, don't expect much. For that price you can bet much changing and customizing will be required. And why not? The standard professional business packages sometimes go for over \$2000 for one package! It's time to separate the hobby world from the business world. When a businessman buys a system, he's told to expect to pay anywhere from 1/2 of the system price to 2 times the price for software alone!! So let's don't be naive and expect \$20 software to perform like the \$2000 package!

Dean Gibson puts it a good way. When you go to the store to look at a product that you're not sure about, do you decide to STEAL it and try it out? Software piracy has too nice a name. The bottom line is STEALING. Call it ethics, morality, Christianity, Brotherly love or whatever. If you've once been burnt, then the lesson is, let the buyer beware. I'm making a big deal out of this because I now know personally of several cases being tried where the 'pirate' is being sued for many thousands of dollars for 'trading' software. It just isn't worth it. By the way, how do the authors feel?

Dean Gibson says if he has strong enough evidence that his 400K SY.DVD driver has been 'pirated', when HDOS 3.0 comes along, guess who will be the only one having a 400K driver available for it? (Dean of course!). Enough said.

MICROSOFT ROYALTIES

Bill Rosen sends me in some information with respect to royalties due to Microsoft when selling programs compiled using the Microsoft compiler. From Microsoft. "The distribution of applications compiled using Microsoft's compilers does not require any licensing or payment of royalties to Microsoft. The one exception to this policy is the BASIC Compiler."

"Beginning with Version 5.30 of BASCOM, two run-time libraries are distributed with the compiler. OBSLIB is the run time library normally used to compile systems consisting of a single program since compiling with OSLIB will cause a copy of that library to be linked in with every program compiled. .4BRUN is the common run-time module first implemented in version 5.30 of BASCOM. BRUN contains the most common library routines, and can

be used by any number of compiled programs thereby avoiding duplication of the run-time on disk for applications requiring several modules. . . Distribution of the BRUN module does require you execute the license and payment of \$40.00 royalty to Microsoft for every computer you license for BRUN." For more information on this, write MICROSOFT, INC./ 10800 NE Eighth Suite 819/ Bellevue, WA 98004/ (206) 455-8080.

H89 COUSIN TO H8'S PC-12

Don Long, of Environmental Control Systems/ 9319 Willowview Ln/ Houston, TX 77080/ (713) 464-9315, says they will shortly be releasing a similar card for the H89 as the PC-12 is to the H8. As you may know, the PC-12 gives you 12 parallel I/O ports, plus a real time clock (RTC). The PC-89 has 9 bidirectional parallel ports with perpetual clock calendar which plugs into the left side of the H89. It will provide the user with the means of connecting a multitude of devices to the system without jeopardizing the integrity of the system by extending the bus outside the machine or loading the bus (same concept as with the PC-12 for the H8). Prices and availability to be announced in the next issue of this newsletter.

ADDITIONAL INFO FROM TRIONYX

Trionyx is also introducing a ground connection kit for their T-H90 motherboard for the H8. It will allow the pc boards to be grounded to the motherboard without having to install a second connector on the pc boards. It has been designed to make the T-H90 motherboard to be a more attractive proposition.

The effectiveness of the new motherboard in solving a wide range of system related problems has now been demonstrated beyond any doubt (I'll vouch for that--ed). The partially assembled Heath equivalent version of the T-H90 motherboard is available for \$175, in addition to the fully assembled version at \$250. It is important to note, that the T-H90 is an exact replacement for the original Heath H8 motherboard and no modifications are required to use the Heath H8 printed circuit boards with the Trionyx motherboard. The addition of ground wires or additional connectors to the pc boards is completely optional.

The Ground connector kit for the T-H90 motherboard sells for \$10 per kit. Each kit contains sufficient material to add ground connections to as many as 6 printed circuit boards. It includes 12 individual pin connector assemblies (insulated pin connectors with wires attached), 1 25 pin gold connector strip to cut single pins from which install at motherboard ground locations, and complete instructions.

This GC-TH90 ground connector kit provides an extremely simple and very effective method of grounding the printed circuit boards which plug into the T-H90 motherboard of the H8. An average of 2 ground connectors will be used for each printed circuit board.

The GC-TH90 is used in place of the T-H90 PC Board Connector Expansion Kits when the motherboard "B" connectors are used only to provide ground connections for the pc boards.

About their new H8 Z80 CPU board. Trionyx has decided to supply their own front panel monitor firmware for their new Z80 CPU. It will be sold separately for \$20. However, the Z80 board will work with any Heath front panel ROM. This firmware package has been developed by Ultimeth Corporation and has been specified to meet the Trionyx standards for quality and performance.

REMARK ISSUE 18 SIDE SELECT

I mentioned in a previous issue that the side select circuit presented in REMark issue 18 may not work. Here is the dope on that. Some have contacted me saying it works

fine, others saying it works not at all, and still others say it works sometimes. Knowing the individuals having trouble, I know they are not likely to be having problems because of improper construction. The only conclusion I can come to is the circuit, for some reason, is marginal, and depending on parts, layout, and installation it will sometimes work and sometimes not.

CP/M BIOS 80 BUG

There has been a minor bug found in the CP/M BIOS 80 which was distributed for the 400K system. It is something that may never show up, but the author Ray, doesn't want something out there with a bug. If you have ordered this before November 1, 1981, return your distribution diskette to whomever you ordered it from for a free update.

UPDATE ON LUCIDATA PASCAL FROM POLYBYTES

Polybytes/ 325 19th St. SE/ Cedar Rapids, IA 52403 is making available to their customers a fresh revision to the version 2.8 compiler/ run-time system. Level I of the Lucidata Pascal system offers access to the 8080/Z80 I/O space, and support for binary constants in Pascal source programs. A slight improvement in convenience at a very moderate cost; \$2.50 with a disk exchange or \$5 without. They are also offering complete support to the HA-8-3 Color graphics board (including AM9511) for \$30 (\$105 for those unlucky few who haven't discovered them yet). Write for details or send money for full doc and two disks full.

A Bug has been found in the Z80 run-time system. If you are trying to read REAL numbers from a disk FILE OF CHAR, you have probably discovered some strange error message like ?02 Channel is not open. This is our way of telling you that a pointer got clobbered and is no longer with us. The problem is easy to fix, but somewhat awkward to PATCH, so Polybytes recommends that you order the latest revision to the system. If you really don't want Level I, send a disk and the'll return a repaired PRUNZ80. Overseas customers please add \$4 to the above prices.

FILES FOR MICROSOFT'S M80

For \$7, or \$2 if you furnish disk, David Green/ 2930 D Columbiana Ct/ Birmingham, AL 35216 is offering his ACM like files for the Microsoft M80 macro assembler. He can handle the 5" in 40TK SS SD or 80TK DS DD. The following are included and are roughly equivalent to the files supplied with HDOS 2.0. HOSEQÜ, HOSDEF, ROMSUB, ECDEF, EDCON, EDVEC, ESVAL, EDRAM, ESINT, OVLDEF, MTR, ASCII. These all have the .MAC extension. Also TEST.MAC for those who have M80 with "include" capability, this tests the previous 'decks'. TEST.SYM is a result of assembly of TEST.MAC for comparison. HOSACM.DOC is the documentation for these files. JUMBLE.MAC is a program from REMark recoded for HDOS and M80 which helps solve jumble puzzles.

Dave also sends along a tip for those who want to verify if the disk format is still valid in all files. Do a COPY ND:=SYN:*.*/*S. PIP will read all files and flag the first unreadable file on the console.

AUTOSCRIBE 2.0?

Almost. I read the TECHNICAL FORUM on page 6 of H8SCOOP, October 1981 and said to myself--Self, Hmmm (I do read what I print!). If you can create a customized bootable disk, you should be able to uncustomize a customized disk! I was right. I was able to get AutoScribe up and running under 2.0, but some problems existed. I will throw this out for those interested, to play with. If you come up with anything, let me know. I finally abandoned the project because of the precious time element.

First, make an AutoScribe disk and remove flag on SYSCMD.SYS if locked (it should not be). Keep in mind,

this AutoScribe is under HDOS 1.6. Next RENAME GO.ABS=SYSCMD.ABS. Now make a minimum HDOS 2.0 disk, and copy all files from AutoScribe disk to 2.0 disk. You'll find you now have AutoScribe and still full control of the operating system. By typing GO you get into the AutoScribe loop, and execution is as normal. The only problem I have had is when you delete a file, it is indeed deleted from the directory, but you still get a disk full message from AutoScribe. Evidently it has a built in Directory that doesn't get updated or something. That's a start, see where it takes you.

NEW STUFF

DISCOUNT ON SUPERCALC

Lum Buckles/ 410 Townsend St/ San Francisco, CA 94107 has a special deal on the CP/M SUPERCALC. It is being sold for \$200 (regular \$295) while supplies last.

I have been told to beware of this program. If you have more than one type of disc system, you must purchase it for the one that you want to use, as you may not be able to copy it to any other disc or system. I guess they want to be sure it won't be copied, and as a result, the end user may be limited to what he can do with it.

MORE FROM SOFTWARE TOOLWORKS

The Software Toolworks/ 14478 Glorietta Dr/ Sherman Oaks, CA 91423/ (213) 986-4885 has two new software products. ELIZA an artificial intelligence program makes you wonder if it's really a computer. Eliza carries on a conversation with you, in plain English. You can play it or program it, using any text editor to modify the script or create completely new ones. Comes with the complete original DOCTOR script which converses like a psychiatrist, plus an improved DOCTOR and full instructions for script writing. Requires 40K, specify 8" CP/M or 5" dual HDOS-CP/M format. Price is \$24.95.

SPELL is a program to check your documents for accurate spelling. It lists misspelled words and optionally creates a document copy with errors marked. Works with most word processors including WordStar and Magic Wand. The 20,000 word root dictionary is effectively multiplied many times by a prefix/ suffix table, yet packed by sophisticated algorithms into fewer than 30,000 bytes. It can look up over 3,000 words per minute at 4Mhz, or 1,500 at 2Mhz. You can add new words and customize. Compact 16,000 word dictionary also provided for 48K machines, which is the minimum memory required. This too is available on 5" HDOS or 5" and 8" CP/M. These are available at most Heath stores or from Software Toolworks. Price is \$49.95. Add \$2 per order postage and handling, \$3 overseas, 8" disks \$3 or \$5 overseas. California residents add 6% sales tax.

They also have a C Compiler that generates MACRO-80 Code. C/80 is an affordable compiler for the C programming language and is now available in a new 1.6 release providing random access file I/O and Macro-80 compatibility. It generates fast 8080 assembly language or Microsoft's Macro-80 relocatable assembler and linking loader.

C/80 supports all C control statement and arithmetic and logical operators, character and 16-bit integer data, pointers, arrays, strings, data initialization, in-line assembly language, conditional compilation and most other preprocessor functions. It lacks structures, pointers to pointers, and long floating data types.

The runtime library provides standard C I/O functions, and such large system features as dynamic storage allocation, I/O redirection, and a runtime execution profile facility for easy program optimization. The cost is \$39.95 plus \$3 shipping for 8" CP/M disk, or \$2 shipping for 5"

Heath/Zenith CP/M or HDOS disk. Registered owners of previous releases may update for \$10 plus shipping.

FROM REICHERT DIGITAL SYSTEMS

Roy Reichert/ 29 Blazier Rd/ Warren, NJ 07060 tells me the LOGIC program mentioned in issue #19 sells for \$24.95 and not \$29.95 as reported. There is also now a CP/M version of XYLOT/ BARLOT for those interested for the same price as the HDOS version, \$29.95. They should be available from many Heath stores.

They have also added PERMDEX, a program designed to produce a permuted index for any listing of titles, reference, catalog entries, key-words, etc. Provides a valuable cross reference for easy look-up. It sounds like a similar concept to KEY-IT mentioned a few issues back. Requires 56K RAM, and at least 2 drives, and is supplied on a dual HDOS-CP/M disk.

FMS, (Financial Management System) designed for use in managing home finances tracks bills and payments on charge accounts, etc., and maintains a monthly check register. You can set up a system of up to 25 accounts for which you normally make regular payments. FMS will automatically track these accounts and tell you when any given payment is coming due. FMS will print out any account report or check register, as well as the tax-deductible expense record. There are more features, write for the catalog. Price is \$39.95, requires 48K RAM and 2 disk drives.

ADDRESS for \$24.95 is a unique mailing address system. It permits selective filing of addresses by attribute, and uses random file I/O for up to 200 addresses. Entries may be sorted by last name or Zip code. In cases where name spelling is uncertain, it is not necessary to provide an exact match guess for the name. The SOUNDEX code feature permits you to enter any phonetic spelling which approximates the desired name. Requires 56K RAM, 1 disk drive, and a printer. For more information on these or their other software products, write for a catalog, or call (201) 469-3854.

UPDATE ON 8" DISK CONTROLLERS

Ray Livingston of Livingston Logic Labs/ 2386-323 E Del Mar Blvd/ Pasadena, CA 91107 wants the following information passed on. Refer to October H8SCOOP page 4 for the complete story. The controllers for the H8 and H89 are now ready. I feel this is the best cost effective way to get 8" single density for CP/M usage. It will allow you to take any standard 8" CP/M program disk and run them on your H8 or H89. The idea behind this controller is a very low cost reliable system for CP/M compatibility. There have been some changes since I last mentioned this in October.

Installation no longer requires cutting any traces on the CPU board. It does not support 5" drives which doesn't matter since the Heath controller can be used along with the 8" controller with no additional changes. A replacement I/O decode ROM is no longer supplied as it is not required. The existing ROM can be used. This was done to ensure compatibility with any ROM changes Heath may make in the future.

Because of higher production costs, the controller prices have been increased. The bare board plus software for the H8 is \$114.95, and \$99.95 for the H89. The kit with all parts is \$217.95 for the H8 and \$199.95 for the H89. The assembled and tested controllers are \$249.95 for the H8 and \$224.95 for the H89.

The Keyboard Studio (address further on) will be handling these products. Ray passes along this information concerning these 8" controllers. Up to 4 drives, either single or double sided in any combination may be used of the 8" variety while still retaining your 3 5" drives under the Heath controller. Each single sided 8" drive stores

250K while the double sided stores 500K. The minimum requirements for H8 are 32K RAM, extended configuration board or Heath's Z80 board, any Z80 CPU (will not run on 8080). The complete kit for H8 is \$217.95, while the assembled and tested version is \$249.95. For the H89, you must also have 32K RAM, and the ORG 0 ROM mod. Kit prices are \$199.95 while assembled and tested version is \$224.95.

Full software support for both are provided for your CP/M operating system. Modifications to the Heath BIOS are done automatically with a supplied program, and new MOVCPM, CONFIGUR, and FORMAT are provided as well as Bootstrap loader and Bios loader. These are all supplied on the 5.25" discs. HDOS operating system drivers can be purchased separately.

The H89 card is about the size of the 16K expansion board. The H8 card is half the size of a standard board, and all bus signals are brought across to another set of connectors on the far side, allowing other half-boards to be plugged in, which will be available in the future. Complete documentation including schematic, parts list, source code for all supplied software, assembly and installation instructions are provided. A new monitor ROM is also available which allows you to boot from any device connected to the disk controller.

ON TANDON CP/M BIOS

Ray also mentions some features of his CP/M Bios modification to allow the Tandon drives to be used with CP/M. The package now for \$30 from H8SCOOP will contain the following features. Several source files containing only those portions of the BIOS which require modification. A .SUB file which controls the CP/M ED editor, ASM assembler and DDT debugger to automatically generate the source code for the modified BIOS and re-assemble it to generate the final relocatable BIOS module. Several source files containing those portions of MOVCPM which require modifications to permit generation of bootable systems for both 40 and 80 track disk formats. A .SUB file which automatically modifies the existing MOVCPM5 to generate the new MOVCPM. TINIT, used for formatting both 40 track single sided and 80 track double sided disks. CONFIG, a replacement for CONFIGUR which allows alteration of all system parameters. The complete source code is supplied for all supplied software.

In addition to this, there will be a read-only device driver which allows reading of 40 TK discs in an 80 TK drive. Support will be provided when Heath releases their new CP/M 2.37 to allow you to use the modified BIOS with this release also.

FINANCE FROM BRADLEY

BRADLEY COMPUTING/ POB 242/ Dunedin, FL 33528 offers FINANCE which is a business and investment oriented program. It computes present values, future values, and payment amounts and is intended for use in comparing investment and financing alternatives. It is menu driven, conversational and easily used by the computer novice. It retains previously calculated or entered values as default entries for the current calculation.

FINANCE runs under HDOS with MBASIC and sells for \$19.95 including printed documentation.

MORE FROM KEYBOARD STUDIO

The Keyboard Studio/ 125 Aspen/ Birmingham, MI 48009/ (313) 645-5365 has some new stuff worthy of mention. Caverns of the Doomed for \$22.95 is a real time adventure game with moving graphics and optional voice output for those who have a Vortex Type 'N' Talk. Two versions are supplied on disk, the non talking one requires 48K while the talker requires 56K, running under HDOS MBASIC. The object of the game is to get as many treasures as

possible, but while wandering from any one of 30 caverns you encounter various monsters whose main objectives are to get rid of you. You could even fall into a pit or run into other obstacles. Since this action packed game takes awhile to get through all 30 caverns, provisions have been made to save a game in progress.

COMPACTA for \$24.95 (add \$10 for source code) is a utility written for those having 80 track dual sided 5" drives or an 8" system. It stores an entire diskful of files into a single large file so no space is wasted (see last month's H8SCOOP). Compacted files retain all their directory information internally and may be restored later if desired. Files may be restored on an individual basis or in groups using HDOS wildcards. An alphabetically sorted CATALOG listing may be displayed for a compacted file without having to unpack the file first.

For those who own the Type 'N' Talk, they also have a customized device driver named TNT, complete with interfacing diagrams and doc. The driver has settable options that directly relate to Vortex's capabilities. Also included are assembly language demo programs complete with source code. One called SPELL-IT is a simple CAI program for different levels of spelling skills. The driver is \$19.95. The Type 'N' Talk is also available for \$339 which includes this driver. Add \$1.75 shipping on all orders under \$100.

Ray also mentions that The Keyboard Studio now has in operation a dial-up system accessible after normal business hours. The purpose of the SOFTalk bulletin board system is for callers to be able to gain additional information about products. They can also run demo programs of many of the most recent offerings. There will be special unadvertised reduced prices on items for people using this system.

MICRO MEDIA MAGAZINE

Micro Media Magazine, a floppy disk-based publication for the H19, H89 or Z89 is available from MMM/ Dept 8/ POB 402286/ Garland TX 75040. This is an information source and software source type of publication combined. What makes this different from such as this newsletter is it is supplied on disk and not paper. It is for the most part menu driven. Each issue usually contains unique software applications, lots of graphics, advertising, reports and more. Write MMM for their free software catalog and order form.

STOCK PORTFOLIO MANAGEMENT

From Robert Koestler/ 640 Trephanny Ln/ Wayne, PA 19087 (215) 687-3194 comes an HDOS MBASIC stock portfolio management program. It is characterized by simplicity of operation and yet having versatility on a professional level. Program options are **Create Portfolio, Add Stocks, Delete Stocks, Display and Correct, Sort Portfolio, and Print Report**. Its main function is to print out a complete one page formatted report on entry of the Dow Jones Average for the day and the price of each stock. The report which requires a 132 column page, consists of the company name, number of shares, bases price, current price, dividend, yield, investment, current value, gain or loss, total investment, total dividends predicted per year at the current rate, total value and total gain or loss. If only an 80 column printer is available a limited report can be printed. Many other features are provided.

If you are not into stocks, you can still create any other portfolio you like and follow its progress each day. This program requires H89 or H8/H17 with 48K and comes on 5" disk for \$20. Write Robert to order or for more information.

MORE GAMES FROM EVERYWARE

EVERYWARE/ POB 60802/ Sunnyvale, CA 94088 is offering

two new action graphic games for the H8/H19 and H89 systems. First they want to note that the INVASION and MISSILE CONTROL listed in the August issue of H8SCOOP are now available on dual format HDOS CP/M disks. They are \$17.50 each and require 32K.

The two new games, GALACTIC WARRIOR and Y-WING FIGHTER are fast action machine language graphic games, and are versions of popular video arcade games. You are the GALACTIC WARRIOR on a solo mission thru space. Your mission is to engage the fleet of enemy star cruisers and destroy their space station. Test your skill as wave after wave of interceptors are dispatched from the space station to challenge you. Use your long range missiles to break up the attack formation and your short range laser if an enemy fighter gets too close. Activate your protective force field to shield against enemy missiles, star bombs and delta rays. As you come closer to the space station, the most skilled interceptors are dispatched. You must blast thru the space station hull and make a direct hit on its central reactor.

In Y-WING FIGHTER you pilot a fighter over the mountains of an unexplored planet. You must seek out and destroy the dreaded enemy's home base. Flying just above the ground, you will encounter a variety of unknown, and increasingly more challenging dangers to test your skill to its maximum. You are shown a side view of your fighter and the terrain as it passes below while you control your speed, altitude and powerful front and rear phasers. If you hit, you can survive by gliding your crippled fighter to a safe landing. After repairs are made, you can take off to resume the journey.

Both of these are available for \$19.50 each in dual format 5" disks, and require 48K. They are available from EVRYWARE or most Heath stores.

MORE FROM HOYLE & HOYLE

A new space exploration game "A Galactic Experience", is available from Hoyle & Hoyle Software/ 716 Elam Av/ Greensboro, NC 27403, or at most Heathkit stores. It's a graphic adventure, word logic game with 5 planets to explore and a mystery to solve. More than "hunt and drop treasures," this game assumes you are a Captain in the Galactic Reserve Space Force, recalled for active duty on a special assignment. As a pilot and computer technician, your expertise is essential to the survival of the Galactic Outpost. You must be on guard, as portions of Earth are a radioactive wasteland. Someone on the Galactic Grand Central Space Station is trying desperately to kill you and you will be puzzled while your wisdom is being tested. Requires HDOS, with H19 or H89 terminals, and minimum of 32K. Sells for \$25, but is on sale for \$20 until December 1st. Hoyle & Hoyle will ship COD by UPS or any bank cards. Call (919) 378-1050 and ask for Janet to order.

UPDATES FROM XTRA-SOFT

Xtra Soft Inc/ POB 91063/ Louisville, KY 40291/ (502) 499-1533 has updated their Point of Sale and Inventory packages to System II. System II includes the most demanded enhancements as well as much streamlining and refining to satisfy their stringent criteria. Some update points include the following:

More storage. System II will operate on any disk arrangement and allow up to 32,640 unique items. Larger item numbers have been implemented from the previous 9 character limit to a 2 character CLASS field and a 17 character ITEM number field. The price of the System II Point of Sale is \$399.95, and the System II Inventory Management sells for \$399.95.

In addition to this, they have a business system demo disk for \$7 which is loaded with info and sample processing for both packages. Demo by menu selection or cycle for over 30 minutes. The electronic Cash Drawer is \$359.95. They have some Educational systems software and will be

handling TCS Accounting system, consisting of General Ledger, Payroll, Accounts Receivable, and Accounts Payable. I'm not sure if this will be HDOS or CP/M, write them to order or for more information on any of the listed.

EPSON MX-80 GRAPHICS SUPPORT

Aaron Reynolds of PowerR Co/ 12624 Lakeland Av SW/ Tacoma, WA 98498 (206) 588-2946 announces new software to support the MX-80 with the GRAFTRAX-80 Bit Graphics option under HDOS.

They have a program which allows H19 graphics (also H89) to be drawn on an Epson MX-80 with the GRAFTRAX PROM set installed. The program, called GRAF89 is compatible with the output of many of the graphics editors such as Ed-A-Sketch. They also have available two HDOS device drivers available for the MX-80 with GRAFTRAX thanks to FBE Research.

GRAF89 will not do inverse video alphabetic or numeric characters, it will print them in regular video (it will do the inverse video space through solid block). Because the MX-80 cannot move the paper backwards, pictures which require the cursor to be moved up, cannot be drawn (GRAF89 prints an error message). Many of the HDOS device drivers will not work with the Graphice mode because it requires all 8 bits of the character to be sent (and most HDOS drivers strip the 8th bit (see TECHNICAL FORUM)).

A summary of the products are as listed. GRAF89 for \$15 prints H19 pictures on the MX-80 if the GRAFTRAX PROM is installed in the printer. MXGRAF for \$13 is an HDOS device driver for the MX-80 with GRAFTRAX running on the FBE H89CTI Model B Parallel printer interface. SERMX for \$13 is the same as MXGRAF but for use on the H89 serial I/O card. This also requires the addition of the 2K buffer serial I/O card with the MX-80. GRAF89/SERMX for \$24 includes GRAF89 and SERMX for a savings of \$4.

All orders are shipped on 5" standard nonbootable HDOS disk. Include \$2 shipping per order, WA residents add 5.3% sales tax.

BIBLIOGRAPHY UPDATE

This is not only the latest update, but since I am ending 1981 I am also going to throw in my comments. BYTE magazine is getting larger all the time, and the majority of the space is being taken by advertisements! Even the articles are not printed straight through, but are cluttered with numerous unwanted adds. I wish they, and others, would print the articles first, with all the adds in the back. In the case of BYTE, you'd see a few pages of articles, and a few hundred pages of adds. CREATIVE COMPUTING would be much better off changing their name to APPLE/PET/TRS Gazette, as that's the main content. Dr. Dobbs Journal is much too high priced for much too little. My subscription to this one will not be renewed (neither will CC, and BYTE is questionable). As usual, Microcomputing pulls through again through 1981 with lot's of good stuff on the Heath line, despite Wayne Green's stiff necked attitude about these computers--very ironic!

October 1981 Microcomputing starts out with a cover picture of the dual H17 and H19 terminal. On page 130 "Cool Down the H14" tells how to build a fan box to place under the H14 to keep it cool to print longer without having the high temp light come on and slow down printing. "A Printer for the H89" on page 144 describes an interface for the H89 and the TRS-80 line printer, which is the Centronics parallel printer. Page 162 begins an article by Ray Massa of The Keyboard Studio on "The Heath/Phone Hookup". This article talks about turning the H8 into an inexpensive sophisticated automatic phone dialer, the same thing Ray sells through the Keyboard Studio.

November Microcomputing has an article on page 80 "Made to Order Business Forms" which was written with the H89 in mind. It talks about a way to design professional looking order forms, which utilizes the graphics of the H89. Another article somewhat related is on page 52 "Focus In on Your Financial Picture". Although programmed in MSI Extended Disk BASIC, it uses an H19 terminal for special effects, and should be easy to modify for MBASIC or BH EXTENDED BASIC. Finally on page 225 is a short item "Speed Up your H8 Programs". This is not an article, but a news release on a BASEX compiler for cassette tape systems and the H8 computer. It sounds good, and I hope it becomes available for disk. It is available for \$33 from Interactive Microware, INC/ POB 771/ State College, PA 16801. The blurb says BASEX programs run up to 10 times faster than similar BASIC programs, and program overhead is only 2K bytes for the execution routine.

My comments on the upcoming SEXTANT from Charlie Floto of BUSS? Mixed emotions. I feel they are trying to take the place of REMark which already exists and has been trying (and indeed is doing) to do a better job. I seriously doubt if our limited industry can support both REMark and SEXTANT, and would sure hate to see REMark go, but with the high pay SEXTANT is offering for articles vs REMark's nothing, I wonder? I feel REMark will have to start paying premium to attract GOOD timely and technical articles, or either sink or remain strictly a beginners rag. Anyway, only time will tell. I have been preparing some articles to publish, which would not be appropriate for H8SCOOP and was thinking of Microcomputing, but with SEXTANT's better paid rates, I may go with them!--ed.

REQUESTS

Howard McPike/ 123 Windy Circle/ Brandon, FL 33511 is interested in the game of Contact Bridge. He would appreciate any information on this game in software for the disk operating system. He has the H17 2 disk system.

WHO'S WHO

Art Petkosek writes in that the Northeastern Ohio Heath User Group (NOHUG) meets the second and fourth Thursday each month from 7-9PM at the Maple Heights Public Library (Cleveland Suburb). They have 40 members currently and are growing. Persons desiring information about the club and/or membership should contact Art Petkosek/ 4705 Tanglewood Pl/ Lorain, OH 44053.

Tom Tcimpidiš writes in that a private new HBBS is in operation in the LA area. It is running on an H8 under HDOS. The system supports both public and private messages as well as allowing full access to HDOS and the database. The number is (213) 366-4837 and the system is available most all the time. If the system is available, the phone will be answered by the second ring.

Jim Czebiniak wishes to thank all those people who wrote to help with his MX-80 problems. He ended up buying the Lindley device driver and is very satisfied with it.

MINI REPORTS

BASGEN

BASGEN by James Czebiniak/ 199 Vly Rd. Extension/ Schenectady, NY 12309/ (518) 869-8723 for \$50 is one of the best programs on the market. This was mentioned several times in last years issues of the newsletter, but for those of you who missed it, let me tell you about it. BASGEN is a MBASIC program that allows you to define parameters, variables and protected screens. It automatically generates the MBASIC code for random file I/O handling and protected screens for inputting data. With

this program, you can generate efficient programs such as mailing lists, inventory, or what have you, without having any knowledge of random file I/O operation.

I have used this program to generate an inventory point of sale business system and have sold a complete system running this program which has been extensively modified. The generated code of BASGEN and the generous documentation supplied has taught me much about random file operation and protected screens. Protected screens are something that's hard to describe you have to see it to appreciate it. Anyone who studies a BASGEN generated program and knows something about MBASIC can tailor the program for their own applications (such as with my inventory point of sale program) to end up with a professional program taking minimum space. My inventory point of sale program coupled with XTRA SOFT's cash register made a true business store system which would operate under MBASIC's reset function with only one drive and still handle over 1000 inventory items! We finally went to two drives to make backups etc. easier, but it's a fantastic program and I highly recommend it to all. Write James for more information on this, or to order. I may in the future make my inventory program available. Incidentally, the new BASGEN release is out which supports more functions and is available still for \$50 to new purchasers, or updates to previous purchasers for 3 disks (if you're registered).

H8/H89 SOUND BOARDS

Mako Data Products/ 1441-B N. Red Gum/ Anaheim, CA 92806/ (714) 632-8583 has available the sound boards for the H8 and H89 computers. In one word, FANTASTIC! The H89 board for \$125 plus \$5 handling contains 2 multi channel PSG chips and plugs into P504 and uses any decoded port address. It must be used in the right hand bus of the H89 to take advantage of its full features. It has two AY3-8910 PSG chips, with each making a parallel I/O port available, in addition to the boards features.

The H8 version for \$225 plus \$5 handling features 4 PSG chips with I/O ports available and plugs into the H8 buss, and is switch selectable for I/O port. I have seen them both, and they are both high quality boards with built in amplifier and speaker, capable of external amplifiers and speakers. Both boards have software included for sound demonstrations such as bells, wind chimes, sirens, wierd sounds, music and more. From Assembly or BASIC or any other language allowing you to output to a port, you can set the PSG chips and program almost any type of sound into them including music. Oh yes, they both come fully assembled and tested.

An excellent program available for use with these boards is MICRO PIANO for \$24.95 which displays a graphic picture (for H19 or H89) of a keyboard which you can actually play via keyboard keys to make polyphonic music over an 8 octave range, while creating your own sound envelopes. It's something you have to see and play with to beleive. Excellent support software for an excellent product. As I have more time in the future (I hope), I may do a detailed report on this. Stay tuned to H8SCOOP for the SCOOP!

GREAT GAME--MUNCHKIN

I usually don't have the time to use the computers to play games anymore, but after hearing a few good things about MUNCHKIN, I had to buy a copy and try it. It is one of the first games that I really enjoyed (despite the frustrations) and kept coming back to get them baddies! I swear those munchkins have eyes! This game is one that a wide variety of age groups can appreciate and play with no complications. I don't care for the games were one has to study lenghtly manuals and continually refer back to them to figure out what to do. I also don't go for adventure type games, not because I'm not the adventurous type, but because I don't have the time for them. Munchkin is short and simple. Anyone having a computer and not having

Munchkin, has missed the boat. A few munchkins mix well with business for recreational purposes, you understand. Anyway, in case you don't know, MUNCHKIN is available for \$19.95 from The Software Toolworks/ 14478 Glorietta Dr/ Sherman Oaks, CA 91423/ (213) 986-4885, or from most Heath stores. When the Munchkins come to get you, tell them Henry sent you, and they may give you a break. But watch out--THEY DO HAVE EYES!

TECHNICAL FORUM

RECOVERING BAD DISCS

I imagine many of you have lost some valuable discs for one reason or another, and have been unable to recover them. Here are some helpful hints. If you are unable to mount the disc, chances are the header has been destroyed. The method I found to work in most cases is to fool HDOS into thinking it is OK. What you have to do, is initialize a new disc with the same volume number and label as the destroyed one. Then mount it. Once it is mounted, the system has accepted it, so remove it and insert the bad one. In most cases you will be able to proceed normally from this point.

There were also some good suggestions in the September 1981 issue of CHUG, the Capital Heath Users' Group. Subscription rates to this local HUG newsletter are \$12/year from CHUG/POB 341/Fairfax, VA 22030. Below is reprinted from this issue under "Recovering HDOS files from "Bad Disks".

"There are now several generally available utility programs which will convert HDOS files to CP/M format and vice versa. Two examples are HOSCOPY/ CPMCOPY by Jack Thompson and HDOS2CPM/ H8COPY from HUG.

"Beyond the basic function of converting files between the two systems, I have found these programs to be very useful in recovering HDOS files from disks which are not MOUNTable by HDOS. In most cases I have been able to quickly and reliably recover HDOS files by simply attempting to transfer them to a CP/M disk and then back to another HDOS disk.

"Assuming that sufficient correct information exists in DIRECT.SYS and GRT.SYS, files can usually be recovered without resorting to reading them at the track/ sector level.

"Another technique which sometimes works is to determine the bad sector and to read and rewrite the sector (without altering the data) using the HUG DUMP program or equivalent. This causes the sector control information (headers, checksums, etc.) to be rewritten."

In another article in the same issue, Larry Henderson explains another method to fix a damaged disk using a DUMP type program. Some failures can't be fixed, but most can be repaired by this program. This method assumes you have two or more drives, and if not, change the instructions from SY1: to SY0:.

"Some problems cannot be fixed, those resulting from a hard disk problem such as a scratch, or loss of the DIRECT file by overwriting are almost hopeless. I have found that these types of losses amount to only about 10% of the errors found.

"I will assume that you have a bootable disk mounted on SY0:, and that you have placed and mounted the "corrupt" disk on SY1:. If you can't mount the "corrupt" disk on SY1:, it will be necessary to access the track and sector as listed at the end of the article, with the disk in SY1: and unmounted. Place a piece of paper and writing instrument beside the system to allow note taking.

"Now let's find out where the problem is on the disk. In all cases note which track and sector is being printed on the

screen so that corrective action may be taken at the appropriate area. The most likely area where problems lie is the RGT, so type: DUMP SY1:RGT.SYS.

"If you get a print out without a lot of banging or scraping, the problem probably is not here. If you get a comment about a corrupt disk, the problem is probably in the DIRECT file. If the latter is true, using the location table at the end of this article, list out the individual sectors of the DIRECT for the version with which the disk was initialized. If there is banging and scraping, type CNTRL C to get back to HDOS and proceed with the following general fixit method.

GENERAL FIXIT METHOD

The user inputs will be printed in **BOLD print**, and assumes you are using DUMP.

DUMP (return). Type in the disk (SY1:), Track and Sector (recorded from your previous printout). The system may then grate and groan, looking up what should be on the disk. If it finally reads the sector it will ask if you want to change it. Respond **Y** (return). Enter **FF** for the Byte to change (return), and **RETURN** again. The sector, in decoded format will again be printed on your screen. The system will ask if you want to write the sector to disk. Respond **Y** (return). Again read this particular sector. If there is no bump and grind you have fixed that sector and can go on to bigger and better things.

"If you still get the noise, try the general fixit method several more times. Sometimes it's just a speck of dirt, which may become dislodged with extra tries. If this doesn't work, you probably have lost the use of the disk, unless the error is on track zero.

"Alright, now we have a good RGT file. Now let's look at track 0, sector 9 using the General Fixit Routine. If you found an error and fixed it, or if the sector was ok, proceed to the next technique. If you couldn't fix the problem here, you still have hope. In this case proceed to the TRACK ZERO FIX.

DIRECTORY FILE FIX

"Get back to HDOS with a CNTRL C, and list out DIRECT.SYS by typing **DUMP SY1:DIRECT.SYS** (return). Note the track and sector numbers as they print, and in particular, note any sectors which get groaning and grating prior to printing on the screen. Use the General Fixit Routine to repair any sectors you find in the directory. Repeat this section on the directory until no errors (grinding and groaning) are found.

"If the problem is found to be in loading a particular program or file from disk, you may use the above technique to locate the problem within the file, and then fix it.

"The track zero fix follows this technique and will likely be required to fix any problems with track zero. It seems to me that most errors on track zero have been caused by the track, or a portion thereof, being overwritten by an HDOS/ Hardware error. First, if you have gotten this far without finding uncorrectable errors, your chance of success for saving the disk contents is greatly improved. Now check each sector of track zero for error. If you are able to find and fix the errors that's great. If you can't fix the error, all is still not lost. Proceed with the following technique.

TRACK ZERO FIX

"Initialize a usable disk using the same serial number and HDOS version used on the corrupt disk (see first part of this article--ed). Then go back to the condition where you have a bootable disk in SY0:, with DUMP on it. Place the unmounted newly INITED disk in SY1:. We will now, using the following method, write track zero from the good disk to the corrupt disk.

DUMP (RET)
(Drive) 1 (RET) *

(track) 0 (RET)
 (Sector) 0 (RET) Start with zero, and work through sector 9 as you cycle through the method, going back through the process from the return point indicated by * above. The DUMP program will now list out the sector on the screen and ask if you wish to modify it. Answer Y (RET). (BYTE) FF (RET), and RETURN again. The program will again list out the sector on the screen and ask if you want to write in information to the disk. Remove the GOOD disk from SY1: and replace it with the corrupted disk and then type "Y" (RET). The program will now write the information from memory onto the 'corrupted' disk. When complete with the writing operation the program will ask for the new drive. At this point, replace the 'corrupt' disk with the 'good' disk in SY1:. Now proceed to the return point (*) above and continue to copy the remaining nine sectors from the good disk to the corrupt disk.

That's it. These methods should fix almost all of your problems with unmountable, unbootable, or unreadable disks. The following locations are given for HDOS 1.6 and 2.0.

RGT.SYS for 1.5, 1.6, and 2.0 is Track 1 sector 0.
 GRT.SYS for 1.5 and 1.6 is track 23 sector 8, and for 2.0 is track 14 and sector 8. DIRECT.SYS for 1.5 and 1.6 is track 22, sectors 2, 3, 6, 7, 0, 1, 4, 5, 8, 9 and track 23, sectors 2, 3, 6, 7, 0, 1, 4, 5. For HDOS 2.0 it's tracks 13 and 14, same sectors.

MORE ON MX-80 GRAPHICS

We have talked earlier that graphics on the MX-80 is difficult because the 8th bit is used for graphics to the printer. The 8th bit of a word is commonly also used for parity checking, and HDOS software does not use it, thus it strips it off via ANI 177 command. To make the MX-80 work with graphics, one must go into the device driver .ASM and remove this ANI 177 command where the parity is stripped, and re-assemble the code. This is not a job for the inexperienced assembly programmer. However, once this is done, all 8 bits are outputted to the port. Now if the Epson is set for 8 bits, and the driver is set for all 8 bits (8 bit word), there should be no problem.

Now from Lloyd Richards. "The Heath supplied LPH14 and LPH24 Device Drivers both send 8 bit words, one stop bit and no parity to the MX-80 printer. Set the option switches in the MX-80 for this configuration. (See H8SCOOP issues 13,16,17--ed).

"The OUT command in BHBASIC can be used to send graphics to the MX-80. There are a few things to consider when using the OUT command. The OUT command does not use the device drivers but instead communicates directly with the output port. One of the jobs of the Device Driver is to properly set up the output port to communicate with the printer (called initialization--ed). Therefore something must be sent via the Device Driver to the printer before the OUT command will work. This output can be in your BASIC program or do a CAT LP:= in HDOS. This need only be done once after BOOT and not each time you wish to send something to the printer.

"Another way to send graphics to the printer is with the CHR\$() command in BHBASIC. However this function strips off the eighth bit that is required by the MX-80 for graphics. The CHR\$ function sends an ASCII character but the MX-80 graphics are outside the ASCII range of 0 to 127D.

"This can be corrected using a modified patch program. (SEE REMARK ISS #17 page 18 if you have not modified patch). There are two addresses that must be patched in BHBASIC." Change address 057130 from 346 to 000 and 057131 from 177 to 000. (Lo and Behold, 346 is the ANI instruction, and 177 is the and mask for stripping the 8th bit!--ed). This will allow the CHR\$() to send any word in the range 0 to 255D, which now includes the graphics set. Whether you use the OUT or CHR\$ commands, a CR must

be the last character sent to the printer to cause it to print the contents of its buffer.

Besides this, or if you are using MBASIC, try one of the special device drivers previously mentioned in H8SCOOP.

ON H8 FRONT PANEL

D.C. Shoemaker writes in "The H8 front panel is the single most versatile part of the system, and the part most frequently ignored by the user. Most folks seem to take the approach that the lights look swell, but leave it at that. Hopefully, the following information will make it a bit easier to do the useful things Heath intended when they incorporated the panel.

"There are two memory locations at the start of RAM that PAM8 uses to control the front panel, and the contents of those addresses determines what you see on the LED's. The first is 8199D which governs whether PAM8 displays a memory location and its contents, or a register pair and their contents. In BASIC, POKE 8199,1 to display a register and POKE 8199,2 to display memory. If you choose to display a register pair, POKE 8197,N to select the register pair, choosing the value of N from the following:"

If N=0, Register SP will be displayed. If N=2, AF. If N=4, BC. If N=6, DE. If N=8, HL. Lastly, if N=10 Register PC will be displayed.

HEATH UCSD PASCAL ON TANDONS

Since the Heath UCSD Pascal package does not run under HDOS, it is impossible to use it with any other drives besides the standard 100K drives it was designed for, since it does all it's own I/O. This is something I must caution readers against. When buying software, either HDOS or CP/M, be sure it operates under one of the systems if you are likely to ever expand and it may not support the extra components. This is the case with UCSD which will not recognize any other drives than the old 100K drives. Another example, although removed from this, is AutoScribe. Although it runs under HDOS, it looks like they may have shelved the updates, and thus, it also will not support any other drives than the 100K. I hate to say it, but it looks like Heath is dropping lot of HDOS support in favor of CP/M since they don't have to rewrite all the software for HDOS to make it go! That will be the subject of a future editorial. Getting back to the point of this subject, if you have UCSD Pascal from Heath and wish to use your Tandons, all is not lost. This article will tell you how to use the Tandons as 100K drives--better than nothing!

This tip comes from Rick Comito/ 62 Haviland Av/ Lynn, MA 01902. Bootable discs for the Tandons to work as 100K discs can be prepared as follows.

- 1) INITialize a disk in the Tandon Drive. (A media check is recommended first).
- 2) Configure the plug-in jumpers in your Siemens or Wangco drive as SY0: (all of this is much easier if you have changed the jumpers to DIP switches--a practice all should follow).
- 3) Configure the Tandon as SY1:.
- 4) Boot PASCAL up in the standard 100K SY0:.
- 5) With the disk prepared in step 1 in the Tandon, type T for T(ransfer, and when the prompt 'Transfer what file?' comes up, type '#4:;#5:'. 6) EX(ecute) Booter. When the prompt 'What is the source disk (0,4,5,9)' appears, type '4'. When the prompt 'What is the destination disk (0,4,5,9)' appears, type '5'.
- 7) Reconfigure the Tandon as SY0:, and the Siemens standard 100K as SY1:.
- 8) RESET the computer and BOOT.

Of course you can prepare as many bootable disks as you like at one time as long as they have all been initialized in the Tandon with the INIT program. Pascal's Format Utility won't do it for you in a Tandon. Unfortunately, the Tandon looks like a SS 40TK drive to UCSD Pascal, but at

least you can use the drive.

4MHZ H89 MOD

For the daring here's a simple mod to the H89 to cause it to operate at 4Mhz. This was made available by Patric Harvey/ Square Wave Info Sys Inc/ 1233 SW Goldcoast/ Palm Bay, FL 32905/ (305) 725-1562.

On the CPU board of the H89, U502, a 74LS92 is used to divide down the clock. Pins 9 or 11 have 4MHz on them. What is needed is to cause the CPU and memory to see 4MHz while the rest of the board sees 2Mhz.

To accomplish this, on the REAR (Not the parts side) side of the CPU, cut the trace leading to pin 3 of U512. Connect 4MHz from pin 9 or 11 of U502 to this pin (pin 3 of U512). You may want to put a SPDT switch in so you can switch between 2 and 4 MHz operations. This hardware mod connects 4MHz to the CPU, the single step logic, and the memories.

You may have to swap out the Z80 for a Z80A, although I have not had any problems. If you got an early H/Z89, you may have memory parts which are already fast enough to run 4Mhz. If your machine crashes often at the new speed, you probably will need to change them. A similar mod can be made on the CRT board, since the clock divider circuitry is almost identical, and also, a jumper is provided for 3.1Mhz operation. At 4Mhz the screen will run at 19.2K with no trouble. At 3.1Mhz it isn't fast enough to keep up. The Static RAMs on the CRT board (2114's and 2112's) can't deal with 4Mhz, they should be replaced with 250 ns parts. Everything else should handle it OK. I'd advise to do one change at a time. I don't think the CRT is needed as much as the Z80 speed mod--ed.

Besides these hardware mods, you will also have to make some software changes. This one for HDOS operating system. You must change the contents of the H17 disk controller ROM as follows. 037.145 from 24 to 50 (split octal). 037.146 from 24 to 50, and 037.147 from 120 to 265. This patch works for HDOS 1.6 and 2.0 for sure. I have found these corresponding locations (I think--I have not had the time to try this yet) in RAM for HDOS 2.0. 040.123, .124, and .125 respectively have these values in them. I would think one could boot at 2Mhz, then reset and change these values, and then if you have a good switch, very fast flip it to the 4Mhz position, and you should come up. Otherwise, have the ROM burned by Ray Livingston, or for \$18 Rick Martinek of California Computer Consultants/ 5903 W Park Hill/ Milwaukee, WI 53212 (414) 774-8445 will do it.

Besides this, you must also make the CP/M patch if operating under CP/M. In the file BIOS.ASM (supplied with CP/M by Heath) change the following equates (values given in Decimal). WHNA from 20 to 40, WHDA from 20 to 40, and WSCA to 181 (old value is 64*25/20). Then prepare your BIOS as described in the Heath manual using PREL. Finally patch your BIOS preloader, which resides with CP/M in the first 3 tracks of the disc as follows. With a GETPUT type program (as described in CP/M Manual), load the first 3 tracks of the bootable diskette into memory, starting at 4380H. Then change the following: 5D93 from MVI A,14 to MVI A,28. 5D9E from MVI A,14 to MVI A,28. Finally, 5DBF from MVI A,50 to MVI A,0B5H.

Note the modified boot ROM will work OK for 2 or 4 Mhz, so you don't have to worry. Certain programs under both operating systems do their own disk I/O. These include TEST, TEST17, and many HUG utilities such as DUMPs. Under CP/M, DUP doesn't work. That's where the switch is handy, to change back to 2Mhz if using these. If anybody has done this and has any feedback, please let us all know.

ON DISK FORMATS & CONTROLLERS ETC

There has been a lot of confusion generated of late on double density controllers and disk formats, and I include

myself. The introduction of the Heath Christmas catalog was the last straw. This was a very poorly organized and worded catalog and left most people in total confusion. I want to take this space to tell you what I know about the entire situation, possibly to prevent hasty decisions.

First, let's talk about disk formats. Presently under HDOS with the standard 100K drives, single density and single sided you have 102.4K of storage. The way to figure that out is 10 sectors per track, and 40 tracks at 256 bytes per sector. That gives 10 sectors times 40 tracks or 400 total disk sectors at 256 bytes per sector, or 102400 bytes, which is 102.4K. The next popular format is the Tandon 400K drives which with the standard single density controller, gives 10 sectors at 80 tracks per side, or 160 disk tracks total. 160 tracks times 10 sectors per track is 1600 sectors. Since you are still operating at single density, or 256 bytes per sector, this gives 409.6K. So far so good. You still are using 10 hole hard sector disks and remain totally compatible, which is actually the best and most cost efficient way to go.

Before leaving Heath, we know HDOS 2.0 can't support this card, and neither can CP/M, yet new releases of either were not mentioned in the catalog with the controller board! I heard rumors you may have to send in a card to get the driver software when you get the board, I don't know if it's true, but it sure sounds unorganized!

Now you may ask, if a standard HDOS disk has 400 sectors, why when a CAT is done do only 390 sectors show up? The answer is track 0 is the track with the header and other information, and all 10 sectors are being used for this purpose, and thus are not available to the user, so we only see 390 sectors. Since HDOS assigns sectors in groups or clusters, doing a general CAT may sometimes cause us to "lose" some sectors. The standard 100K drives allocate space in minimum spaces of 2 sectors, so even if a file only takes 1 sector for actual storage, 2 are used up. This can be seen by doing a CAT/ALL from HDOS where actual allocation is shown. The larger the drive capacity, the more the cluster size is. For the 400K Tandon configuration, it is minimum groups of 8 sectors.

As long as we're still on compatibility and disk formats, let's jump to 8" for a moment. There is a CP/M standard, which inefficient as it may be, is indeed the standard. This assumes the 8" drive is set for single side and single density, and is called IBM compatibility. With this you get 128 bytes per sector, and 26 sectors per track with 77 tracks to give 256256 bytes or 256K. Any single sided single density 8" drive for CP/M should be compatible with any 8" controller using this format. That's were the compatibility ends.

Heath's H47 8" SS DD is 256 bytes per sector giving 512512 or 512K (double the number above). Their DS DD (double side double density) again doubles that for 1025024 or 102K, which is about 1 Megabyte. Magnolia's 8" Single side Double density uses 512 bytes per sector, and 16 sectors per track to give 630784 or 630K. Going double sided you again double that for 1261568 or 126K which is 1.2 Megabytes of storage on the 8" drive. It can clearly be seen that the Magnolia double density card for the H89 (no there's not one for the H8) coupled with two 8" drives is better than (and much more reliable) the H47, for less than half the price. If you are simply concerned with CP/M compatibility, than the board from Ray Livingston (see elsewhere in this issue) is the way to go, since it is the lowest cost and is available for both H8 and H89.

What about the CDR card? Yes, it handles 8" and of course has the standard of 128 bytes per sector, and 256 bytes per sector for DD. It looks like they may be compatible all the way with Heath's H47 on this.

Still on Double density, is it actually double? NO. Although it's called double density you don't actually get double storage. The major problem, is because double

density does not mean 20 sectors per track since it is too much cramming, and thus unreliable. When you use the outside surface of the disc, the actual area is much greater than the inside tracks. When on these inside tracks, you are squeezing the information much tighter than on the outside tracks, thus 20 tracks may not be reliable. Read on for the various formats of the various boards.

Talking about 5" drives double density, things really get confusing. There are several options available. CDR Systems and Magnolia Microsystems both came out with a board about the same time for the H89 which would handle double density. Best of all, they handle both 5" and 8" drives, which Heath's card will not do. Unfortunately, especially under CP/M the similarities end. All three of these cards, Magnolia, CDR, and Heath use the Western Digital 1797 Disk controller chip, which introduces some sort of compatibility, but what is done with the format makes the difference. Under CP/M, Magnolia double density uses soft sector disks and stores 512 bytes per sector and 9 sectors per track. This means a standard 40 track drive will get 184320 bytes or 184K Bytes of storage. Quadruple this for the Tandon type drives (double side, double track, and double density, and you get 737280 or 737K bytes of storage. The Heath card uses 256 bytes per sector and 16 sectors per track for 163840 bytes or 163K bytes of storage for the 40 track drive. Using the DD, DT, DS Tandons, you get quadruple this, or 655360 bytes which is 655K bytes. If you note the difference in numbers, you can clearly see you cannot take a Magnolia CP/M disk and expect it to read in a Heath system with the Heath DD controller!

This is too bad. Since Magnolia was one of the first to come out with the format you'd think anyone else following with the DD idea would try to be compatible, right? WRONG! So now we have two systems, both not compatible under CP/M. Enter CDR. Even though they use the same controller chip, it looks like they also won't be compatible. They are using 256 bytes per sector, same as Heath, BUT 18 sectors per track! They also have an extended version which uses 512 bytes per sector at 10 tracks. The first comes to 184K bytes per 40 track, and the extended comes to 204.8K per 40 Tracks. It is reported that it may be marginal, to operate double density at that high a rate, time will tell. The unique thing about the CDR card is a built in hardware stripper circuit which allows one to use 10 hole hard sector discs for soft sector, thus eliminating the need to change to soft sector disks. I would have to question the reliability of using the old disks certified for ST, SS, SD and going soft sector of DD. CDR says they will do all possible to remain compatible with Heath, but again, we'll have to see.

Before I continue, I want to mention some controversy over CP/M directory entries. I am told (and since I'm no CP/M man I don't know for sure--take this for what it's worth) presently Heath supports 64 directory entries with their CP/M, while Magnolia supports 128 for 5", and 192 for 8". Supposedly the new Heath CP/M release will support 256 directory entries and it should be shipping shortly. This is important to consider if you break down entries for accounts etc., where 64 on a disc simply would not be enough.

I'm hoping this is not as hard to read as it is to write. Before I get too far, what about the H8? I feel Magnolia really missed the boat by dismissing the H8. They are leaving out a lot of people who want H8/H89 compatibility for trading software, running on other machines, or perhaps they like myself have both an H8 and H89. I feel they are loosing many thousands of dollars by shelving the H8. Thus far, CDR will support both. Marc Brooks says they expect to have the H8 controller out the first quarter of 1982. Heath said they would have one, but now it looks like they may change their mind (sounds real organized--again). I guess they want to see if the market is there before they go into production.

A small point about the disk controllers for the H8. If the 8080 CPU is to be used, Direct Memory Access will have to be used, and I understand that's exactly what Heath and CDR are going to do. If this will cause problems for some non-Heath memories, remains to be seen.

As far as operating systems, Heath will support HDOS and CP/M always, although it looks as they are slipping very bad on HDOS. Although Magnolia only supports CP/M, Dean Gibson of UltiMeth is releasing a disk driver for that card, so for the H89 anyway, there's HDOS and CP/M. This driver will support the Heath card and Magnolia card, including 8" drives--super! Dean also tells me he will follow the Heath format on the Magnolia board, so any HDOS software should be compatible between Magnolia DD board and Heath DD board. CDR presently supports CP/M, but they are working on HDOS as well.

Is there one more? I guess. Percom now comes out with their card for the H89, but the checking I tried to do revealed little. That worries me. Anyway, they have an HDOS mod for the controller, cables and the DD controller for about \$250. This card works WITH the Heath single density card, so both boards must be in. I guess it's supposed to be able to handle hard or soft sector disks in any drive, software determines what's what and lets the appropriate controller take over. Sounds confusing, but that's what I was told by Charles Bennett who checked this thing out. It also has a parallel port built in. They are supposedly supporting 368.6K per side of a drive, so I would guess that's for the Tandon type drive (double track, double density, double side). They presently will supply HDOS mods to allow it to operate under HDOS, they say CP/M may come. I don't think they're too far on that yet, time will tell how this one turns out.

A short summary. If you may want to expand latter and get the most for your money, forget Heath because they don't support the 8" on their board. I'm not sure about Percom. CDR and Magnolia do support both, and these would seem to be the logical choices. They both do a lot of business, but Magnolia is known more because of more advertising, and a toll free number, which is really a blessing! Let the Buyer Beware!

GOOD GUYS and BAD GUYS

In the October issue I listed BEALS AGENCY as a possible bad guy. I want to follow this up so there are no misunderstandings. It turns out that this was a mail problem and Beals Agency is not to be considered a bad guy. Georges Troupin of Belgium wrote to inform me he had finally received the disk doubler kit which Beals Agency sells. The problem was, it was shipped by surface mail overseas, which took 5 months(!) for Georges to receive!

James Quinn, who is the inventor of this device called me after this was published to try to get things straightend out. He then wrote me a letter explaining several things. The point is, he went out of his way to clear up the matter, and that is the start of a GOOD GUY. Jim also writes "If anyone has a problem with the Disk Doubler or the Beals Agency, they can contact me by mail at: 9120 Clearlake Way/ Lakeside, CA 92040 or by telephone at (714) 561-2540 after 6PM PST and I will personally take care of any problems."

I am grateful to Georges, Jim, and the Beals Agency for clearing up this matter so promptly. I do not want to list anybody as a BAD GUY who really is not. It is easy to see how this type of thing can happen. The most important point is the willingness to clear it up.

GOOD GUY--Dean Gibson of UltiMeth Corp. Craig Chiaverini writes in "I too am very impressed with Dean Gibson's SY device driver. I purchased and installed it without problems but had a few questions on it so wrote to Mr. Gibson. To my amazement he not only got back to me quickly but by phone to boot! He spent over an hour (at

long distance) answering my questions. I came away from that call with a great deal of respect for Mr. Gibson's technical expertise. Saying he has his act together is a gross understatement. I will have NO fears about purchasing any item that has the Gibson name on it."

I have receive similar response from many who have talked to Dean. Indeed he does know his stuff, and although he sells himself as a software consultant, I found he knows a great deal about hardware as well. He has not let me down once, and I know he has helped a lot of others as well. I know we will be seeing more of UltiMeth's products in the future, and like Craig says, we should have no fears about purchasing anything he offers, because he does know his stuff. Hats off to Dean Gibson of UltiMeth for his products and support, above and beyond the call of duty.

BAD GUYS--MARK GORDON COMPUTERS (Again--See issue 16). Robert McClure tells me he ordered some discs from them and his master card was billed but he never received the discs, nor was he able to get any response from them. All he found out is they had their phones disconnected. Stay away from them (if they still exist), since this has happened before. Robert finally got what he wanted from Pacific Exchanges and has been very satisfied with them.

GOOD GUYS--Clark Systems/ POB 490156/ Atlanta, GA 30349. Terje Bolstad tells me their software packages for HDOS and CP/M are very good and well documented. He is using their Mailpro and General Ledger II for HDOS and they work very well for some of their business purposes. He is also recommended for after sales support. Heath's Softstuf's General Ledger, by the way, is Clarks. I have a copy and it is excellent. In fact, I sold an entire system with just this software package to an accountant.

BAD GUYS--Micro Architech/ 96 Dothan St/ Arlington, MA 02174. Terje Bolstad writes in that they sell mainly software for the TRS80, but they also have modified some of the packages for HDOS. Terje purchased their MAIL-H1 for \$79 and it would not even start up properly. They answered his letters, but Terje never did get it running. Micro Architech would neither take it back to fix the problem, nor would they refund his money. I have to vouch for that. I ordered their manual set for all their programs, and they seemed to be the same program modified slightly for different applications, all which seemed to be a general mailing list program. Needless to say, after spending the bucks for the manuals I never ordered the software.

CLASSIFIEDS

FOR SALE--Two WH8-16 RAM boards for \$140 each. One 8K RAM board for \$45. All are in excellent condition. Ted Guest/ 4366 Carlo Dr/ Dayton, OH 45429/ (513) 298-6084.

WANTED--8K memory boards for H8. Will pay \$50 per board and will pay shipping. Will pay or return upon inspection. Must be in excellent condition. FOR SALE--LA36 Decwriter II for \$600. U. S. Energy/ 78-40 164th St/ Flushing, NY 11366/ (212) 380-1004.

FOR SALE--3 Heath 8K RAM boards, factory serviced. \$65 each or all 3 for \$175. Harvy Gack/ 45 Summer St, Apt B-003/ Brockton, MA 02402/ (617) 583-8069.

FOR SALE--Disk drive controller for H89 for \$75. Heath H8 3 port parallel I/O board for \$75. Heath H8 CPU card for \$75. H8 motherboard with tin connectors, ideal for expanding H8 bus for \$25. H8 cassette serial interface board for \$50. H19 disk drive addition parts except cables and Mu Metal Shield for \$20. 3-drive H17 cabinet with power supply and cables for \$225. H8 H17 disk controller card available with this if interested. H8 Wire Wrap Board \$25. Contact H8SCOOP.

FOR SALE--Three 8K Heath RAM boards for H8 for \$50 each. In perfect conditon, contact Joseph McGlone/ 155 Lynn Dr/ Mansfield, OH 44906/ (419) 529-3532 after 5PM.

FOR SALE--H14 Printer with manuals in excellent condition, assembled and burnt in. \$395. To see or for more information, contact Harlow Milham/ 347 Peace Manor/ Palmetto, FL 33561. Phone (813) 722-0629.

FOR SALE--Two WH8-16 16K Static RAM cards for H8, \$200 each. H8-8 8K Heath Static RAM \$80. Contact M. Gulick/ POB 1112/ APO Miami 34001.

FOR SALE--3 H8-1 Static Memory cards (8K), one card missing two 4044 Ic's, otherwise all in perfect condition with manuals. Cards have dip switches installed for fast memory block allocation. 3 for \$100. Robert McCrath/ 519 Revere Dr./ Turnersville, NJ 08012/ (609) 227-6690.

FOR SALE--IBM Selectric I/O typewriter model 735. This is a 10 pitch machine for use with fabric ribbon, 15" platten, 24V solenoids, standard keyboard, in good condition. Requires interface and software (such as H8SCOOP SELECTRIC PROJECT ARTICLE) to make run. \$495 Firm. Stanley Dheim/ 1055 Elm St/ Cleveland, WI 53015/ (414) 693-8760.

FOR SALE--Stretched H8 (16 slots) complete and ready to go, \$200. H8 8K memory boards for \$45 each, Godbout 12K cards for \$65 each. Cassette card \$55, GE tape units at \$25 each. ASR 33 Keyboard Printer with RS232 interface, \$250. ASR 28 \$95. Skintight KSR 28 \$100 (pick-up only on printers); will ship rest UPS at your expense. John Doak/ 45 Allen Drive/ Woodstock, NY 12498.

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PHONE HOURS: 10AM to 9PM as I am available, except sunset Friday to sunset Saturday (Seventh Day Adventist). NO COLLECT CALLS.

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